

MA.912.LT.4.10

Benchmark

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

Benchmark Clarifications:

Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.

Connecting Benchmarks/Horizontal Alignment

Terms from the K-12 Glossary

- MA.912.GR.1
- MA.912.GR.2.6, MA.912.GR.2.7, MA.912.GR.2.8, MA.912.GR.2.9
- MA.912.GR.5

Vertical Alignment

Previous Benchmarks

- MA.7.GR.1
- MA.8.GR.1

Next Benchmarks

- MA.912.LT.4
- MA.912.LT.5

Purpose and Instructional Strategies

In grades 7 and 8, students explored the reasons why some geometric statements concerning angles and polygons are true or false. In Geometry, students learn and construct proofs for many of the geometric facts that they encounter and they learn to use counterexamples to check the validity of arguments and statements. The content of this benchmark is to be used throughout this course. In later courses, students continue to learn to judge the validity of many different arguments and statements.

- Instruction includes the understanding that a valid argument can be a statement or sequence of statements supported by valid reasons, a part of a proof or an entire proof.
- One way to show that an argument is not valid is to provide at least one counterexample to at least one statement in the argument.
 - For example, if the argument is “All rectangles have opposite sides parallel; therefore, given a quadrilateral is not a rectangle, the quadrilateral does not have opposite sides parallel,” then a student can provide a parallelogram as a counterexample to show that concluding statement of the argument is not valid.

Common Misconceptions or Errors

- Students may think a statement is true because they cannot think of a counterexample.

Instructional Tasks

Instructional Task 1 (MTR.3.1)

Part A. Which of the following statements are valid?

- If a quadrilateral is a square, then it is a rectangle.
- All trapezoids are parallelograms.
- Any quadrilateral can be inscribed in a circle.

Part B. Provide counterexamples to prove the invalid statements from Part A are not valid.

Instructional Items

Instructional Item 1

Puaglo said the following statements are valid. Select all the statements that are invalid.

- a. All quadrilaterals have four right angles.
- b. A triangle is a polygon with three sides.
- c. All circles are similar.
- d. All equiangular quadrilaterals are congruent.
- e. A trapezoid must have at least one obtuse angle.

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*